

Kaichen Huang 黄楷宸

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Education

Nanyang Technological University — College of Computing and Data Science 2025.08 – Present

Ph.D. Student · Advisor: Prof. Bo An

Nanjing University — School of Artificial Intelligence

2022.09 – 2025.06

M.Sc. (GPA 4.56/5.0) · LAMDA Group · Advisor: Prof. De-Chuan Zhan

Nanjing University — School of Artificial Intelligence

2018.09 – 2022.06

B.Sc.

Research Interests

World models and interactive / streaming video generation; reinforcement and imitation learning; interpretability of multimodal large language models.

Publications

* Equal contribution.

[1] Runjia Qian*, Zile Wang*, Jihai Zhang*, Kai Zou*, Wei Yu*, Jiaying Li*, Zexiang Liu*, Yaokun Li, Fei Kang, **Kaichen Huang**, Mengyin An, et al.. *Matrix-Game 3.5: Enhancing Real-Time Streaming Interactive World Models with Patch Memory*. [Preprint, 2026](#).

[2] Jiaying Li*, Kai Zou*, Cindy Zhou, **Kaichen Huang**, Junyao Gao, Zile Wang, Yang Liu, Bin Liu, Bo An, Yangguang Li. *DistillAlign: Coordinating Mode Covering and Mode Seeking in Autoregressive Video Distillation*. [Preprint, 2026](#).

[3] Zile Wang*, Zexiang Liu*, Jiaying Li*, **Kaichen Huang***, Baixin Xu*, Fei Kang, Mengyin An, et al., Yangguang Li. *Matrix-Game 3.0: Real-Time and Streaming Interactive World Model with Long-Horizon Memory*. [Preprint, 2026](#).

[4] Jiaying Li*, **Kaichen Huang***, Baixin Xu*, Zexiang Liu, Xianglong He, Zile Wang, Junyao Gao, Yang Liu, Ying He, Bo An, Yangguang Li. *Direct Autoregressive Diffusion Distillation via Error-Aware Causal Pretraining*. [ECCV, 2026](#).

[5] Wenshu Fan, Shenghua Wan, Xinchun Li, Hai-Hang Sun, **Kaichen Huang**, Le Gan, De-Chuan Zhan. *Behavior Cloning Optimization via Secondary Learning*. [CCML, 2025](#).

[6] Shanghai AI Lab (**Kaichen Huang** as contributor). *SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law*. [Preprint, 2025](#).

[7] Yunkai Dang*, **Kaichen Huang***, Jiahao Huo*, Yibo Yan, Sirui Huang, Dongrui Liu, Mengxi Gao, Jie Zhang, Chen Qian, Kun Wang, Yong Liu, Jing Shao, Hui Xiong, Xuming Hu. *Explainable and Interpretable Multimodal Large Language Models: A Comprehensive Survey*. [Under review, TPAMI](#).

[8] **Kaichen Huang***, Shenghua Wan*, Minghao Shao, Shuai Feng, Le Gan, De-Chuan Zhan. *Leveraging Separated World Model for Exploration in Visually Distracted Environments*. [NeurIPS, 2024](#).

[9] **Kaichen Huang**, Jiahao Huo, Yibo Yan, Kun Wang, Yutao Yue, Xuming Hu. *MINER: Mining the Underlying Pattern of Modality-Specific Neurons in Multimodal Large Language Models*. [Preprint, 2024 \(submitted to ICLR 2025\)](#).

[10] **Kaichen Huang***, Minghao Shao*, Shenghua Wan, Hai-Hang Sun, Shuai Feng, Le Gan, De-Chuan Zhan.

SENSOR: Imitate Third-Person Expert's Behaviors via Active Sensoring. [Preprint, 2024](#).

[11] **Kaichen Huang**^{*}, Hai-Hang Sun^{*}, Shenghua Wan, Minghao Shao, Shuai Feng, Le Gan, De-Chuan Zhan. *DIDA: Denoised Imitation Learning based on Domain Adaptation*. [Preprint, 2024](#).

[12] Yihan Liu, **Kaichen Huang**, Yechao Bai. *An Error Analysis Method for Externally Measured Data*. [Measurement and Control Technology, 2023](#).

Experience

Skywork AI · Riemann Dynamics 2025.08 – 2026.10

Joint-Training Ph.D. Student · Mentors: Yangguang Li, Wei Yu

Real-time interactive world models and video generation — Matrix-Game 3.0 / 3.5, DistillAlign, Error Forcing (ECCV 2026).

Shanghai AI Laboratory 2024.11 – 2025.03

Research Intern · Mentor: Jing Shao

Multimodal safety and reasoning — SafeWork-R1.

Hong Kong University of Science and Technology (Guangzhou) 2024.06 – 2024.09

Research Assistant · Mentor: Xuming Hu

Interpretability of multimodal LLMs — MINER, and a survey on explainable & interpretable MLLMs.

Ministry of Industry and Information Technology (MIIT) 2023.06 – 2023.12

Data Processing Engineer

Automated analysis system for industrial economic-trend data; resulted in a granted software patent.

Pengcheng Laboratory 2022.06 – 2023.03

Assistant Engineer · Mentor: Tong Zhang

Medical imaging for congenital heart disease — a ViT / MAE-based ellipse-detection method for robust fetal echocardiography measurement.

Honors & Awards

2023 Academic Scholarship, Nanjing University

2020 First Prize (East China Region), National University Bio-network Design Competition

2019 Third Prize, 9th Nanjing University College Student Career Planning Competition

Academic Service

Conference Reviewer: NeurIPS, ICML, ICLR, AAAI.

Language & Test Scores

IELTS: Overall 7.0 (L 7.5 / R 8.5 / S 5.5 / W 6.5) · GRE: 321 (V 151 / Q 170 / AW 4.5)